

Modality 1:

The embedded mixture of
audio records from two
speakers with additional noise

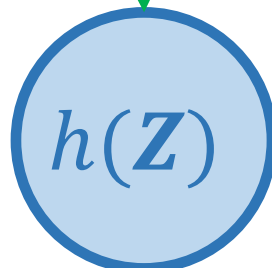
Modality 2:

The embedded video
frames of the cropped lip
region from two speakers

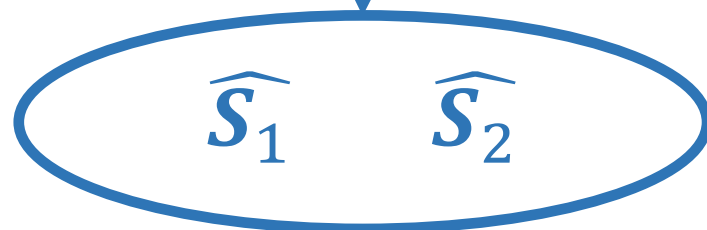
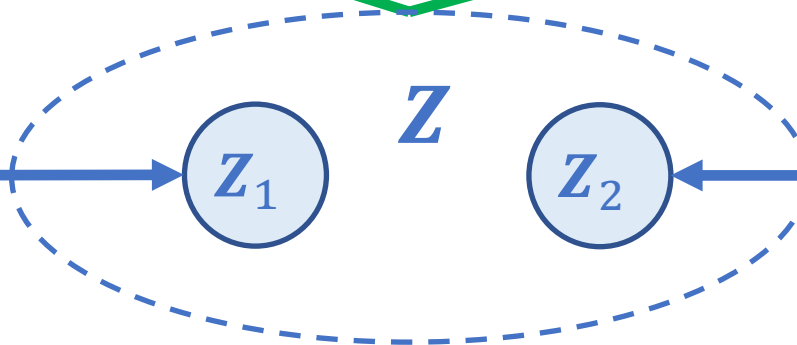
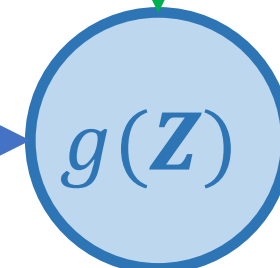
$$S = \text{Embed}(S_1, S_2)$$

$$V = \text{Embed}(V_1, V_2)$$

The audio
latent features
for the two
speakers



The facial
latent features
for the two
speakers



Target:

The separated and denoised audio records
of the two speakers from the mixture of
audio records